

SOFTWARE USER MANUAL

OPC-UA Client Manual V25-1



1. Table of contents

1. Table of contents	2
2. About this document	3
2.1. Legend	3
3. Introduction	3
4. Add and set-up OPC-UA Client	4
4.1. HW settings	4
4.2. Channel setup	6
4.2.1. Channel property mapper	7
4.3. Read/Write channels	7
4.4. Control/Method channels	8
4.5. String channels	8
4.6. Licensing	9
5. Warranty information	10
5.1. Calibration	10
5.2. Support	10
5.3. Service/repair	10
5.4. Restricted Rights	10
5.5. Printing History	11
5.6. Copyright	11
5.7. Trademarks	11
6. Safety instructions	12
6.1. Safety symbols in the manual	12
6.2. General Safety Instructions	12
6.2.1. Environmental Considerations	12
6.2.2. Product End-of-Life Handling	12
6.2.3. System and Components Recycling	12
6.2.4. General safety and hazard warnings for all Dewesoft systems	13
7. Documentation version history	15

2. About this document

The following document will show you how to properly add the OPC-UA-Server plugin in the DewesoftX software and how to properly set all the needed settings.

2.1. Legend

The following symbols and formats will be used throughout the document.



Important

It gives you important information about the subject.
Please read carefully!



Hint

It gives you a hint or provides additional information about a subject.



Example

Give you an example of a specific subject.

3. Introduction

The OPC-UA Client plugin enables DewesoftX to acquire data from any 3rd party OPC-UA server or DewesoftX OPC-UA server. All the data is acquired as asynchronous channels.

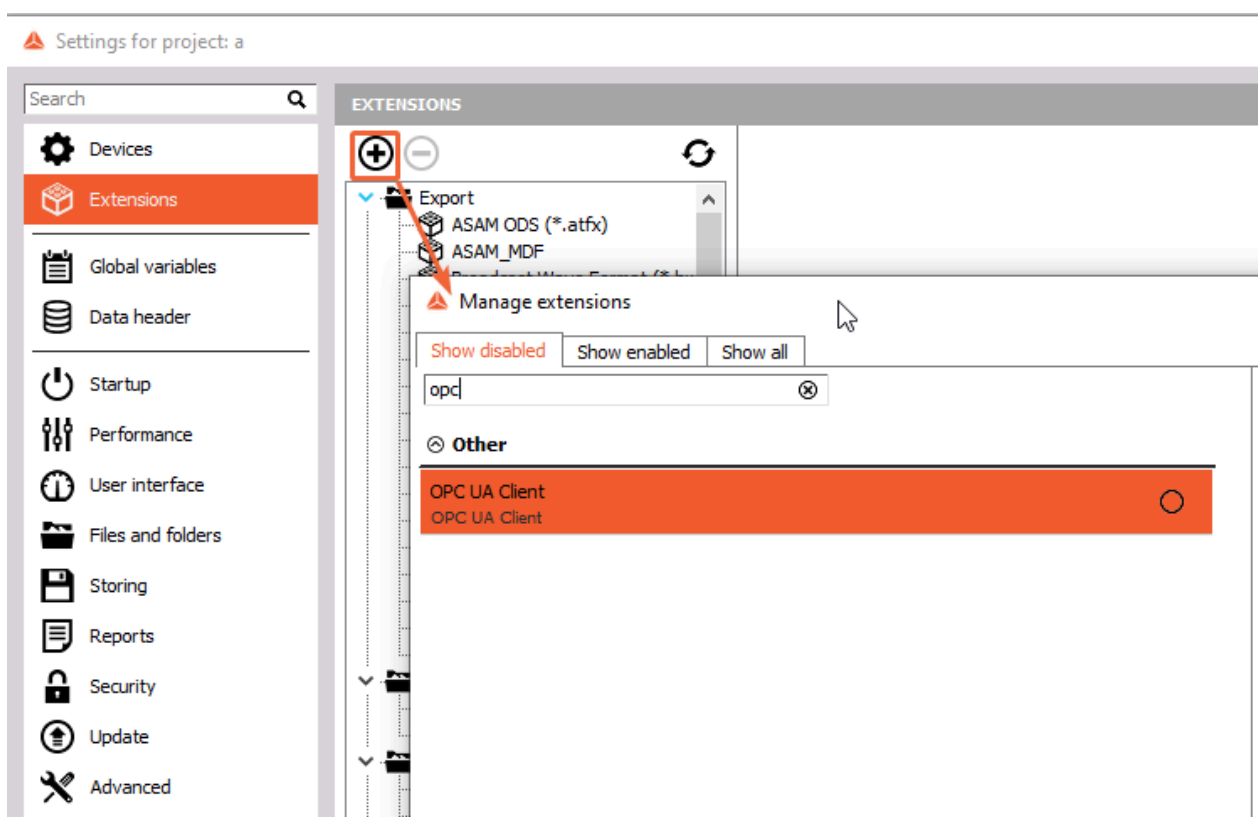
Our OPC-UA Client fully supports subscriptions, which is a standard feature of the OPC-UA protocol. A subscription in the context of OPC-UA refers to a mechanism by which an OPC-UA client can subscribe to receive updates or notifications about changes in data from the OPC-UA server. This allows the client to efficiently receive data without the need for continuous polling, improving performance and reducing network traffic.

4. Add and set-up OPC-UA Client

In this section you will learn how to add the OPC-UA Client plugin in DewesoftX and how to properly set it to your needs.

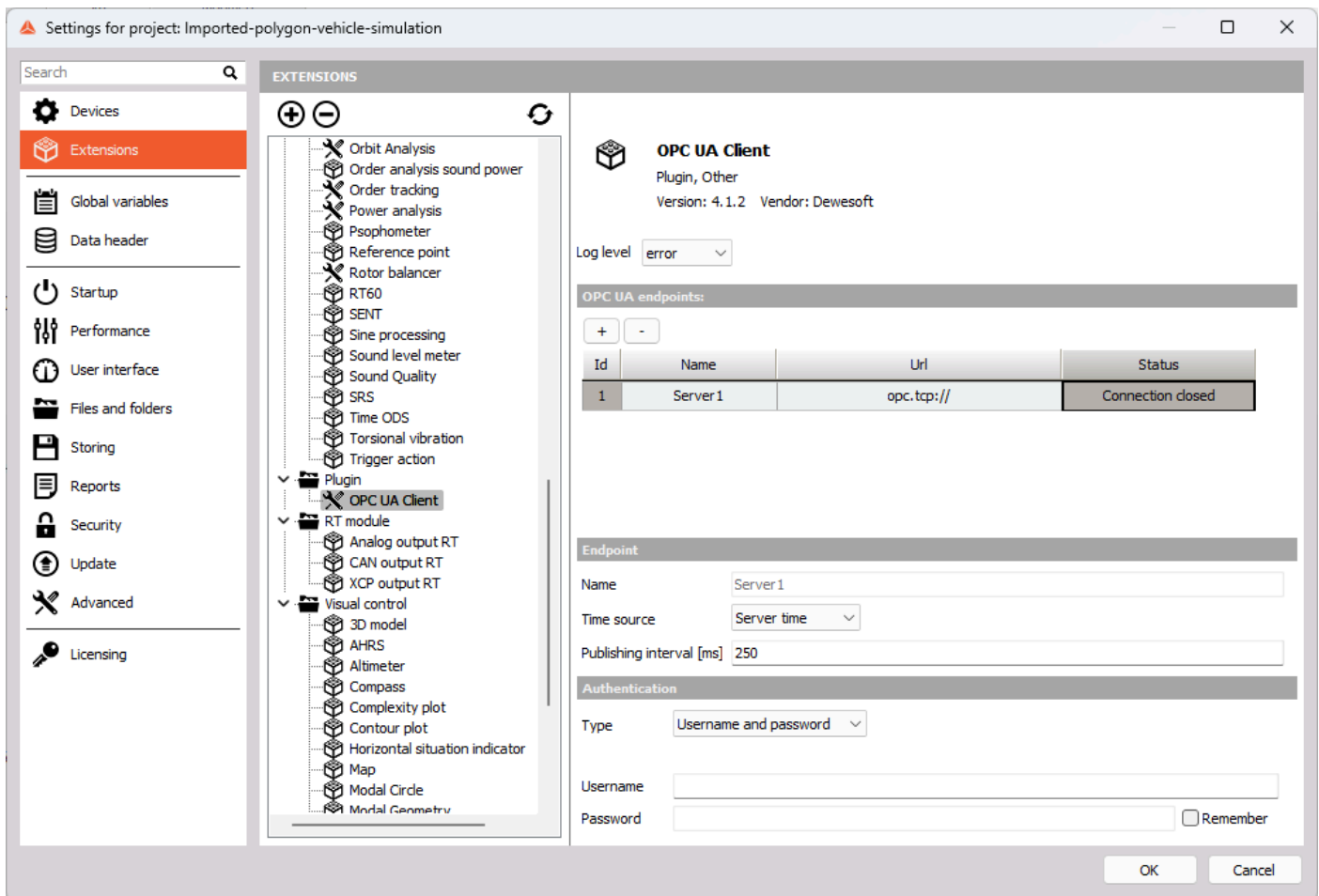
4.1. HW settings

In the HW-settings, under the Extension tab press the plus button. An additional dialog with listed extensions will open. Find the OPC-UA client and enable it. Once the plugin will be enabled it will be listed under plugins.



Adding OPC UA Client plugin

When you click on the OPC-UA client under the listed extensions, an additional dialog will appear where you can adjust the settings for the OPC-UA client plugin.



OPC-UA client settings

On the top of the window the current plugin version will appear. Additionally you can set the following settings.

1. OPC-UA path levels taken for “Ch. no” generation - Ch.no provides additional information of the channel. The user gets the tree-path of the file up to the selected level.
2. Log Level - the level of log that will be saved during the acquisition. The log can also be disabled, by choosing option Off.
3. Define OPC-UA endpoints:
 - a. Users can choose a custom name for the server for a better differentiation.
 - b. The URL of the OPC-UA server need to be entered
 - c. Status will show you if the server is online (available) or offline.
4. Endpoint:
 - a. Time source can be either Server time, which means that DewesoftX will use timestamps that are included in the OPC-UA data package. Or you can choose Local time, which means that DewesoftX will set the timestamp to receive data.
 - b. Publishing interval [ms]: Subscriptions have a publishing interval. The publishing interval of a subscription defines the cyclic rate at which the subscription executes. Each time it executes, it attempts to send a NotificationMessage to the client. NotificationMessages contain notifications that have not yet been reported to the client. This means that the

client will receive data samples from the server in batches at each publishing interval. A publishing interval of 100 ms is recommended. Reducing this value decreases latency but increases the number of requests per second, leading to higher communication overhead. If you choose a longer publishing interval, it is advisable to also increase the calculation delay in Dewesoft settings.

5. Authentication
 - a. Username and password is available.

4.2. Channel setup

The OPC UA Client setup screen allows you to select the Server by clicking the dropdown menu. In the Config tab you can use the browser to browse through the information model. The current location is shown on top of the Browser. You can write the location path and click on **Go** to jump in the location. Under **Nodes** you can see nodes contained at chosen paths along with properties: Identifier, Name, Type and Status of each node. You can select which nodes you want to use in DewesoftX with the Used toggle (Off/On) button.

Used	Index	Id	Name	Type
On	4	Demo.Static.Scalar.BaseD...	BaseDataType	Variant
On	4	Demo.Static.Scalar.Boolean	Boolean	Boolean
Off	4	Demo.Static.Scalar.Byte	Byte	Byte
Off	4	Demo.Static.Scalar.ByteS...	ByteString	ByteString
Off	4	Demo.Static.Scalar.DateTi...	DateTime	DateTime
Off	4	Demo.Static.Scalar.Double	Double	Double
Off	4	Demo.Static.Scalar.Duration	Duration	Duration
On	4	Demo.Static.Scalar.Expan...	ExpandedNodeId	ExpandedNodeId

OPC-UA Client Config tab

In the Channels tab all the selected OPC-UA channels are listed. For each channel you can see its ID, Data Type and current Value. With OPC-UA Client 4.19 version also Array channels are supported.

For each channel you can define the channel color, name and the wanted sample rate. The maximum sample rate also depends on the OPC-UA server settings and capabilities. When you right-click on the Sample rate column you can set the sample rate to maximum - depending on the server limitation. Under the throughput column you will see the actual sampling rate of this channel.

The channels that are added to DewesoftX are always treated as asynchronous data.

Measure

Analyze

Setup files

Ch. setup

Measure

Store

Save

Save as

Storing

Analog in

Real-time

Math

OPC UA Client

OPC UA Server

More...

Remove

OPC UA Servers

Type

Status

Server 1 (opc.tcp://KAL-EVA-S:48020)

connected

Config

Channels

Channels

	+	Color	Index	Id	Name	Type	Sample rate [Hz]	Throughput	Min	Value	Max	Units	Remove
Variables													
			4	Demo.Static.Scalar.BaseD...	BaseDataType	Variant	20.00		0.00		5.00	-	Remove
			4	Demo.Static.Scalar.Boolean	Boolean	Boolean	20.00		0.00		5.00	-	Remove
			4	Demo.Static.Scalar.Byte	Byte	Byte	20.00	0.00	-5.00	AVG 0.00 (-)	5.00	-	Remove
			4	Demo.Static.Scalar.ByteS...	ByteString	ByteString	20.00	0.00	-5.00	P/P NAN / NAN (-)	5.00	-	Remove
			4	Demo.Static.Scalar.DateTi...	DateTime	DateTime	20.00	0.00	-5.00	P/P NAN / NAN (-)	5.00	-	Remove
			4	Demo.Static.Scalar.Double	Double	Double	20.00	0.00	-5.00	AVG 0.00 (-)	5.00	-	Remove
			4	Demo.Static.Scalar.Duration	Duration	Duration	20.00	0.00	-5.00	AVG 0.00 (-)	5.00	-	Remove
			4	Demo.Static.Scalar.Expan...	ExpandedNodeId	Expande...	20.00	0.00	-5.00	P/P NAN / NAN (-)	5.00	-	Remove
			4	Demo.Static.Scalar.Float	Float	Float	20.00	0.00	-5.00	AVG 0.00 (-)	5.00	-	Remove
			4	Demo.Static.Scalar.Guid	Guid	Guid	20.00	0.00	-5.00	P/P NAN / NAN (-)	5.00	-	Remove

Setting max sample rate on OPC-UA Client



Important
Currently only Methods without input arguments are supported.

4.2.1. Channel property mapper

With the OPC-UA 4.19 version we also introduced the automatic channel metadata mapping. This functionality assigns the properties like name, unit and description that are stored in separate nodes to the the CurrentValue channel, which previously contained only raw values.

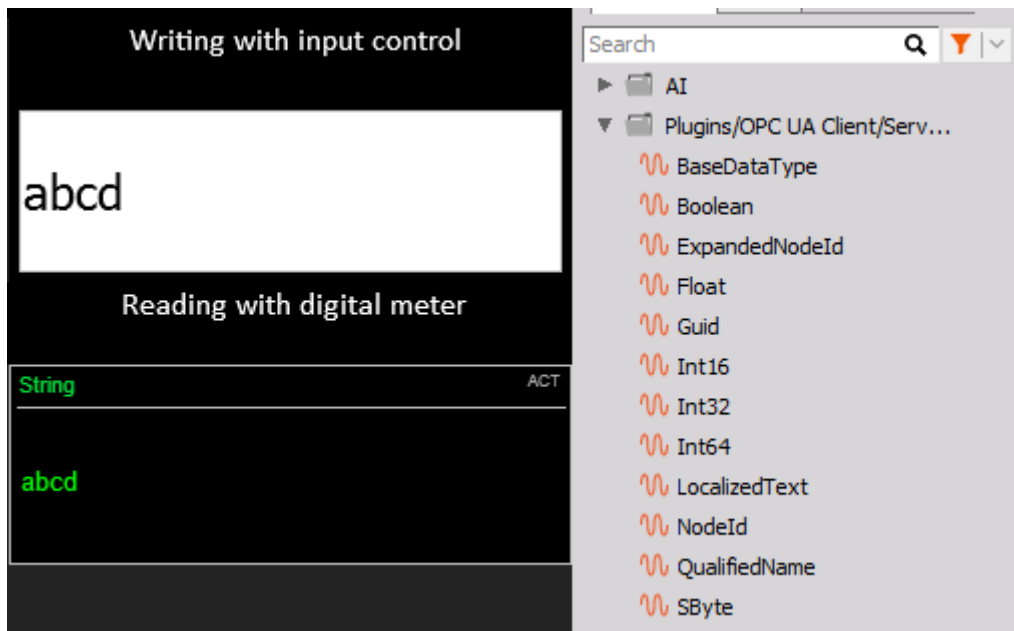
The mapping feature works exclusively between DewesoftX OPC-UA modules, and doesn't apply to external OPC-UA servers, as their structure may vary.

	+	Color	Index	Id	Name	Type	Sample rate [Hz]	Throughput	Interpolation	Min	Value	
Variables												
			1	data/OpcUaServer/channels/7000/0/0/1/channel_CUR_SCA_VAL	AI 1/AmplFFT	Double	100.00	16.00	Auto	0.00	[0.288243, 0.549953, 0.275...	
			1	data/OpcUaServer/channels/1/3/channel_CUR_SCA_VAL	Strain-1	Double	100.00	16.00	Linear	-40000...	AVG 361328.13 (um/m)	400

All channel properties are now mapped to CurrentValue channel

4.3. Read/Write channels

If the node from the OPC-UA server supports Read/Write functionality, you can assign those channels to the input control widget. In this way you can also change the node's value on the server site.

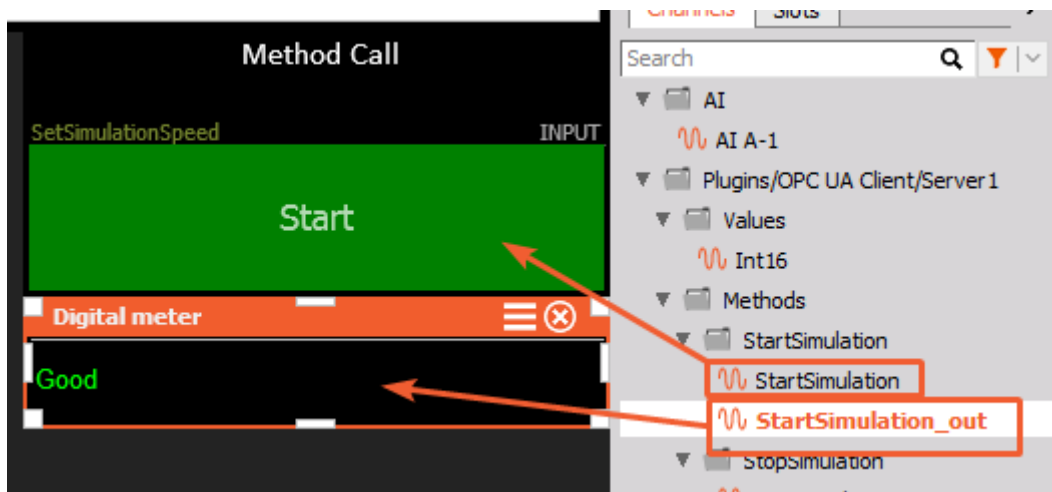


Writing into OPC-UA channel

4.4. Control/Method channels

The OPC-UA Client plugin supports calling OPC-UA methods, without any input arguments. Once you select your Method node in OPC-UA client, you can easily assign it to the input control widget. Use the Push button so you can easily call the methods.

Additionally you have the status channel, so you can check if the method call was successful.



Method Calls from OPC-UA Client

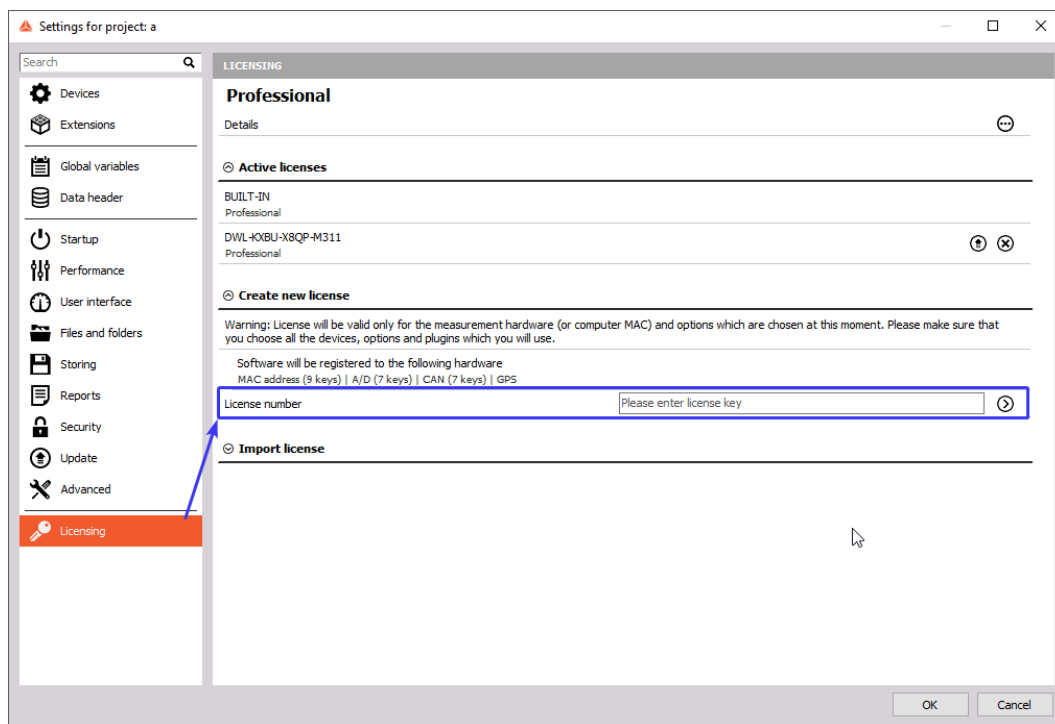
4.5. String channels

String channels update only on value change as opposed to other channels which update on the set rate.

4.6. Licensing

For using OPC-UA client plugin you need a valid DewesoftX license option - DEWESOFT-PLUGIN-OPC-UA-CLIENT. To evaluate the plugin, you can also use an Trial license which you can get from the www.dewesoft.com web-page.

To properly activate the licenses open DewesoftX, go to Options ▢ Settings, select Licensing and expand the Create new license section. Enter the license code and click the Register online button. Then your new license key will show up in the list and have the Status Valid.



Add the license inside DewesoftX

5. Warranty information

Notice

The information contained in this document is subject to change without notice.

Note:

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The copy of the specific warranty terms applicable to your Dewesoft product and replacement parts can be obtained from your local sales and service office. To find a local dealer for your country, please visit <https://dewesoft.com/support/distributors>.

5.1. Calibration

Every instrument needs to be calibrated at regular intervals. The standard norm across nearly every industry is annual calibration. Before your Dewesoft data acquisition system is delivered, it is calibrated. Detailed calibration reports for your Dewesoft system can be requested. We retain them for at least one year, after system delivery.

5.2. Support

Dewesoft has a team of people ready to assist you if you have any questions or any technical difficulties regarding the system. For any support please contact your local distributor first or Dewesoft directly.

Dewesoft d.o.o.
Gabrsko 11a
1420 Trbovlje Slovenia

Europe Tel.: +386 356 25 300
Web: <http://www.dewesoft.com>
Email: Support@dewesoft.com
The telephone hotline is available Monday to Friday from 07:00 to 16:00 CET (GMT +1:00)

5.3. Service/repair

The team of Dewesoft also performs any kinds of repairs to your system to assure a safe and proper operation in the future. For information regarding service and repairs please contact your local distributor first or Dewesoft directly on <https://dewesoft.com/support/rma-service>.

5.4. Restricted Rights

Use Slovenian law for duplication or disclosure. Dewesoft d.o.o. Gabrsko 11a, 1420 Trbovlje, Slovenia / Europe.

5.5. Printing History

Version 2.0.0, Revision 217 Released 2015 Last changed: 23. July 2018 at 16:54.

5.6. Copyright

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5.7. Trademarks

We take pride in our products and we take care that all key products and technologies are registered as trademarks all over the world. The Dewesoft name is a registered trademark. Product families (KRYPTON, SIRIUS, DSI, DS-NET) and technologies (DualCoreADC, SuperCounter, GrandView) are registered trademarks as well. When used as the logo or as part of any graphic material, the registered trademark sign is used as a part of the logo. When used in text representing the company, product or technology name, the ® sign is not used. The Dewesoft triangle logo is a registered trademark but the ® sign is not used in the visual representation of the triangle logo.

6. Safety instructions

Your safety is our primary concern! Please be safe!

6.1. Safety symbols in the manual



Warning

Calls attention to a procedure, practice, or condition that could cause the body injury or death



Caution

Calls attention to a procedure, practice, or condition that could possibly cause damage to equipment or permanent loss of data.

6.2. General Safety Instructions



Warning

The following general safety precautions must be observed during all phases of operation, service, and repair of this product. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of design, manufacture, and intended use of the product. Dewesoft d.o.o. assumes no liability for the customer's failure to comply with these requirements.

All accessories shown in this document are available as an option and will not be shipped as standard parts.

6.2.1. Environmental Considerations

Information about the environmental impact of the product.

6.2.2. Product End-of-Life Handling

Observe the following guidelines when recycling a Dewesoft system:

6.2.3. System and Components Recycling

Production of these components required the extraction and use of natural resources. The substances contained in the system could be harmful to your health and to the environment if the system is improperly handled at its end of life! Please recycle this product in an appropriate way to avoid unnecessary pollution of the environment and to keep natural resources.



This symbol indicates that this system complies with the European Union's requirements according to Directive 2002/96/EC on waste electrical and electronic equipment (WEEE). Please find further information about recycling on the Dewesoft web site www.dewesoft.com



Restriction of Hazardous Substances

This product has been classified as Monitoring and Control equipment and is outside the scope of the 2002/95/EC RoHS Directive. However, we take care of our environment and the product is lead-free.

6.2.4. General safety and hazard warnings for all Dewesoft systems

Safety of the operator and the unit depend on following these rules.

- Use this system under the terms of the specifications only to avoid any possible danger.
- Read your manual before operating the system.
- Observe local laws when using the instrument.
- DO NOT touch internal wiring!
- DO NOT use higher supply voltage than specified!
- Use only original plugs and cables for harnessing.
- You may not connect higher voltages than rated to any connectors.
- The power cable and connector serve as Power-Breaker. The cable must not exceed 3 meters, the disconnect function must be possible without tools.
- Maintenance must be executed by qualified staff only.
- During the use of the system, it might be possible to access other parts of a more comprehensive system. Please read and follow the safety instructions provided in the manuals of all other components regarding warning and security advice for using the system.
- With this product, only use the power cable delivered or defined for the host country.
- DO NOT connect or disconnect sensors, probes or test leads, as these parts are connected to a voltage supply unit.
- Ground the equipment: For Safety Class 1 equipment (equipment having a protective earth terminal), a non-interruptible safety earth ground must be provided from the mains power source to the product input wiring terminals.
- Please note the characteristics and indicators on the system to avoid fire or electric shocks. Before connecting the system, please read the corresponding specifications in the product manual carefully.
- The inputs must not, unless otherwise noted (CATx identification), be connected to the main circuit of category II, III and IV.
- The power cord separates the system from the power supply. Do not block the power cord, since it has to be accessible for the users.
- DO NOT use the system if equipment covers or shields are removed.
- If you assume the system is damaged, get it examined by authorized personnel only.
- Adverse environmental conditions are Moisture or high humidity Dust, flammable gases, fumes or dissolver Thunderstorm or thunderstorm conditions (except assembly PNA) Electrostatic fields, etc.
- The measurement category can be adjusted depending on module configuration.
- Any other use than described above may damage your system and is attended with dangers like short-circuiting, fire or electric shocks.
- The whole system must not be changed, rebuilt or opened.
- DO NOT operate damaged equipment: Whenever it is possible that the safety protection features built into this product have been impaired, either through physical damage, excessive moisture, or any other reason, REMOVE POWER and do not use the product until the safe operation can be verified by service-trained personnel. If necessary, return the product to Dewesoft sales and service office for service and repair to ensure that safety features are maintained.
- If you assume a more riskless use is not provided anymore, the system has to be rendered inoperative and should be protected against inadvertent operation. It is assumed that a more riskless operation is not possible anymore if the system is damaged obviously or causes strange

noises. The system does not work anymore. The system has been exposed to long storage in adverse environments. The system has been exposed to heavy shipment strain.

- Warranty void if damages caused by disregarding this manual. For consequential damages, NO liability will be assumed!
- Warranty void if damage to property or persons caused by improper use or disregarding the safety instructions.
- Unauthorized changing or rebuilding the system is prohibited due to safety and permission reasons (CE).
- Be careful with voltages >25 VAC or >35 VDC! These voltages are already high enough in order to get a perilous electric shock by touching the wiring.
- The product heats during operation. Make sure there is adequate ventilation. Ventilation slots must not be covered!
- Only fuses of the specified type and nominal current may be used. The use of patched fuses is prohibited.
- Prevent using metal bare wires! Risk of short circuit and fire hazard!
- DO NOT use the system before, during or shortly after a thunderstorm (risk of lightning and high energy over-voltage). An advanced range of application under certain conditions is allowed with therefore designed products only. For details please refer to the specifications.
- Make sure that your hands, shoes, clothes, the floor, the system or measuring leads, integrated circuits and so on, are dry.
- DO NOT use the system in rooms with flammable gases, fumes or dust or in adverse environmental conditions.
- Avoid operation in the immediate vicinity of high magnetic or electromagnetic fields, transmitting antennas or high-frequency generators, for exact values please refer to enclosed specifications.
- Use measurement leads or measurement accessories aligned with the specification of the system only. Fire hazard in case of overload!
- Lithium ion batteries are classified as not hazardous when used according to the recommendations of the manufacturer described in Battery Safety Data Sheet, which is available for download from [this link](#).
- Do not switch on the system after transporting it from a cold into a warm room and vice versa. The thereby created condensation may damage your system. Acclimatise the system unpowered to room temperature.
- Do not disassemble the system! There is a high risk of getting a perilous electric shock. Capacitors still might be charged, even if the system has been removed from the power supply.
- The electrical installations and equipment in industrial facilities must be observed by the security regulations and insurance institutions.
- The use of the measuring system in schools and other training facilities must be observed by skilled personnel.
- The measuring systems are not designed for use in humans and animals.
- Please contact a professional if you have doubts about the method of operation, safety or the connection of the system.
- Please be careful with the product. Shocks, hits and dropping it from already- lower level may damage your system.
- Please also consider the detailed technical reference manual as well as the security advice of the connected systems.
- This product has left the factory in safety-related flawlessness and in proper condition. In order to maintain this condition and guarantee safety use, the user has to consider the security advice and warnings in this manual.

EN 61326-3-1:2008

IEC 61326-1 applies to this part of IEC 61326 but is limited to systems and equipment for industrial applications intended to perform safety functions as defined in IEC 61508 with SIL 1-3.

The electromagnetic environments encompassed by this product family standard are industrial, both indoor and outdoor, as described for industrial locations in IEC 61000-6-2 or defined in 3.7 of IEC 61326-1.

Equipment and systems intended for use in other electromagnetic environments, for example, in the process industry or in environments with potentially explosive atmospheres, are excluded from the scope of this product family standard, IEC 61326-3-1.

Devices and systems according to IEC 61508 or IEC 61511 which are considered as “operationally well-tried”, are excluded from the scope of IEC 61326-3-1.

Fire-alarm and safety-alarm systems, intended for the protection of buildings, are excluded from the scope of IEC 61326-3-1.

7. Documentation version history

Version	Date	Notes
1.0	8.6.2018	Initial version
1.1	9.3.2021	Added string channel chapter
V21-1	8.9.2021	Updated to new Template
V21-1	24.1.2022	Updated pictures
V24-1	20.2.2024	New 4.0. Version update
V24-2	13.8.2024	Authentication option
V24-3	24.12.2024	Publishing interval
V25-1	6.2.2025	Support of Array Channel Channel property mapper